

WALMART

Customer Retention Analytics

Power BI Project Report

Video link: <https://www.loom.com/share/a8de2031596c4216ac77ecce7be14bb9>

Introduction & Project Objective

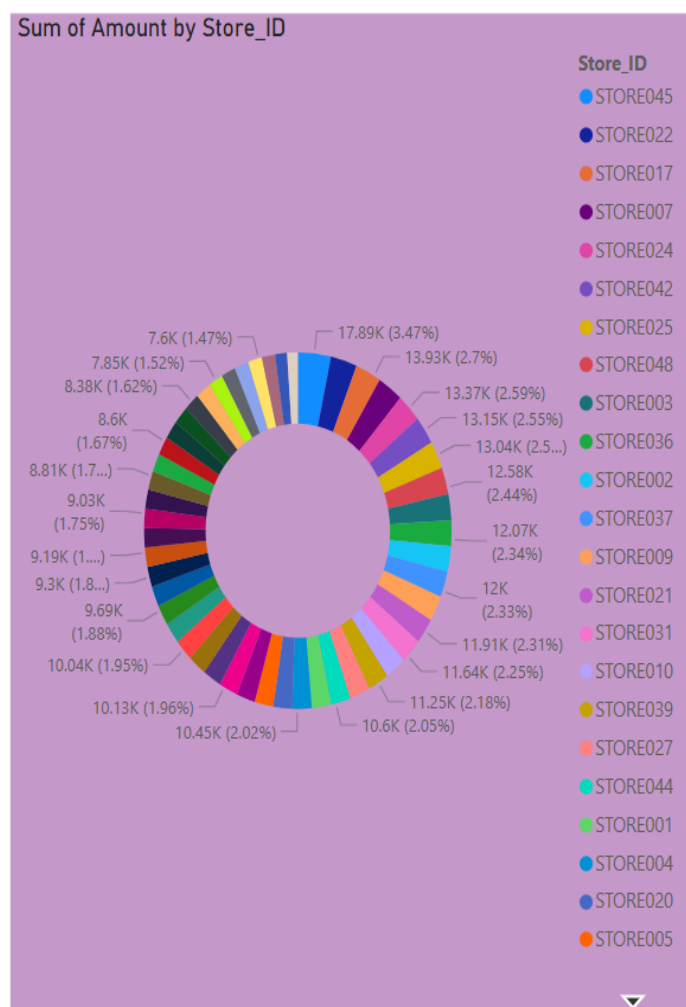
The project uses five interconnected Excel datasets covering 300 Walmart customers. Below is a summary of each table loaded into Power BI:

Table Name	Rows	Key Column	Purpose
Demographics	300	Customer_ID	Customer age, gender, region, income, channel
Transactions	1,000	Transaction_ID	Purchase history, amount, product, store
Churn	300	Customer_ID	Churn flag (0/1) and last purchase date
Loyalty	300	Customer_ID	Loyalty tier, points earned and redeemed
Stores	50	Store_ID	Store type, region, and opening year

This project analyzes customer retention, churn behavior, loyalty engagement, and customer lifetime value using Power BI dashboards. The objective is to identify factors affecting customer retention and provide business recommendations to improve loyalty and reduce churn.

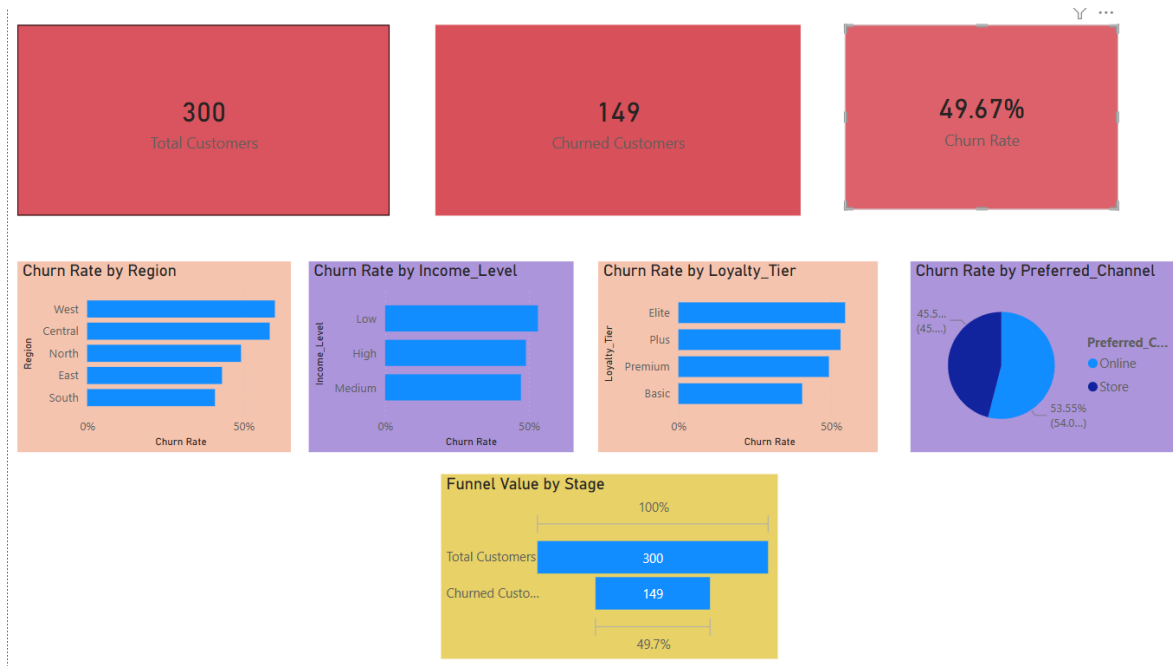
Task 1: Data Modeling & Cleaning

- Data was cleaned by handling missing values and ensuring proper formatting of customer, transaction, and loyalty fields.
- Relationships were established between customer demographics, transactions, and store data to enable accurate analysis.
- Calculated columns and measures such as Churn Rate, CLV, Repeat Rate, and Purchase Segments were created for deeper insights.



Task 2: Churn & Retention Metrics

300 Total Customers	149 Churned Customers	49.67% Churn Rate	50.33% Retention Rate
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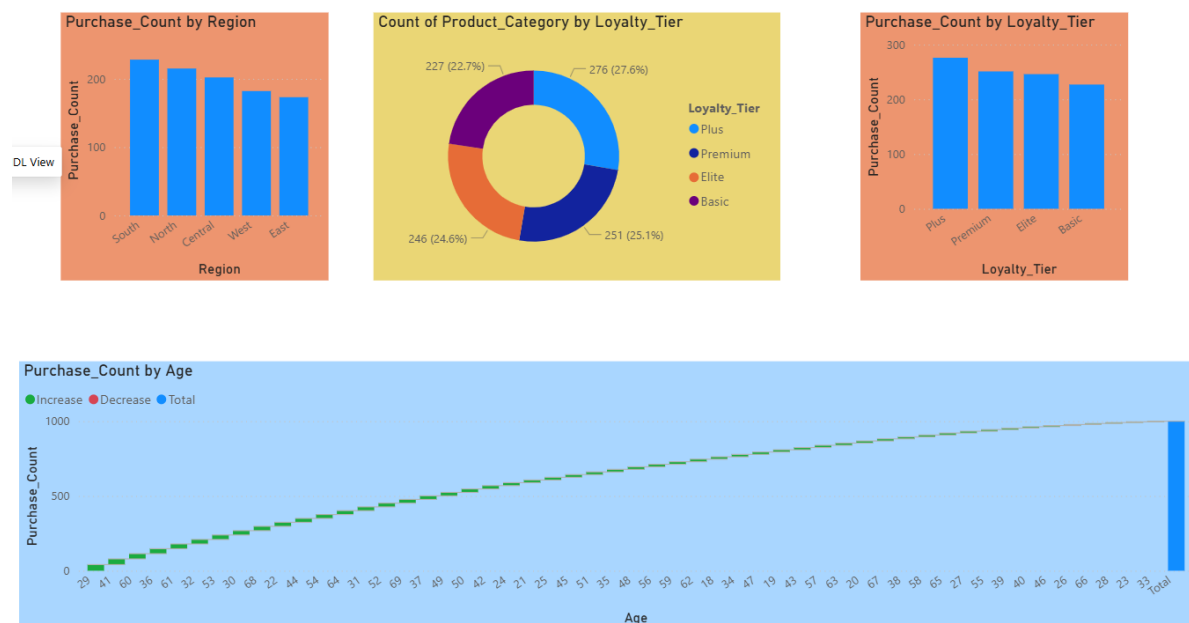


- The overall churn rate is approximately **49.67%**, indicating that nearly half of the customers are inactive or lost.
- Customers from the **West and Central regions** show relatively higher churn compared to other regions.
- Customers belonging to the **Low-Income group** demonstrate the highest churn tendency.
- The **Basic loyalty tier** has lower retention compared to higher-value loyalty groups.
- Customers using the **Store channel** show higher churn compared to Online customers, highlighting weaker in-store engagement.

Task 3: Repeat Purchase Analysis

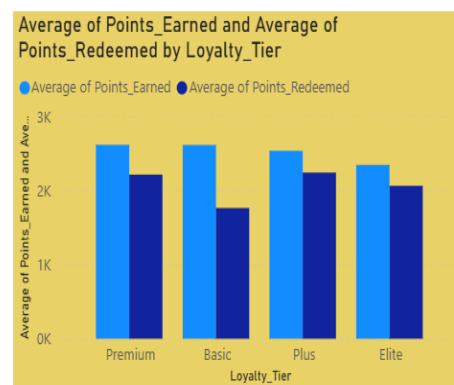
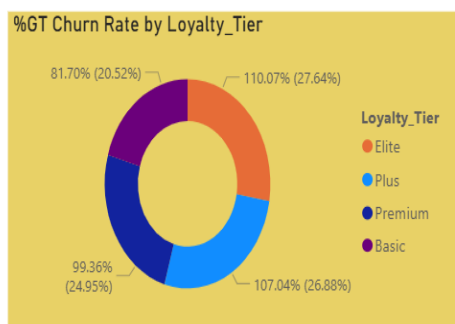
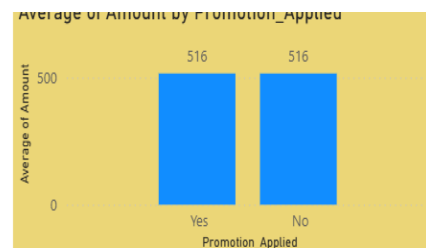
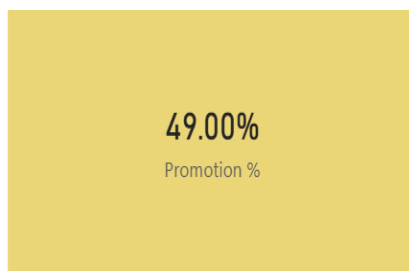
3.4 Avg Purchases / Customer	169 Low-Tier Customers	123 Mid-Tier Customers	2 High-Tier Customers
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Task 4: Promotion & Loyalty Impact

49%	Rs 516	Rs 516	0%
Promotion Rate	Avg Amount (Promo)	Avg Amount (No Promo)	Promo Lift



◆ 1. Promotion Strategy Optimization

Promotions are widely used (~49%), but they do not significantly increase average purchase value. Walmart should implement **targeted and personalized promotions** instead of broad discounts.

◆ 2. Improve High-Tier Customer Retention

Higher churn observed in premium tiers indicates a risk of losing valuable customers. Walmart should introduce **exclusive benefits and personalized engagement strategies** for high-tier members.

◆ 3. Enhance Loyalty Program Engagement

Points earned are higher than points redeemed, showing low engagement. Walmart should **simplify the redemption process and encourage usage through reminders and incentives**.

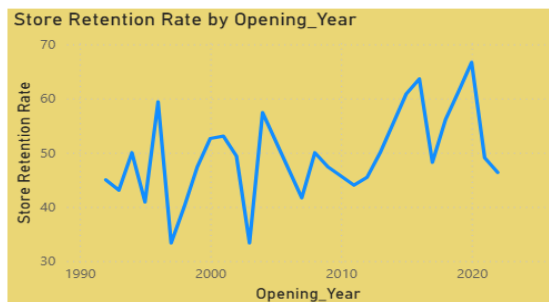
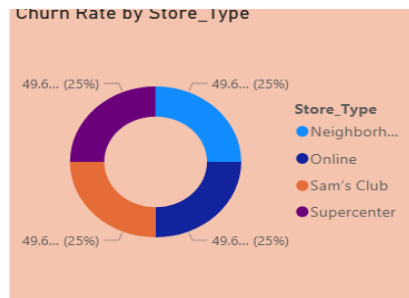
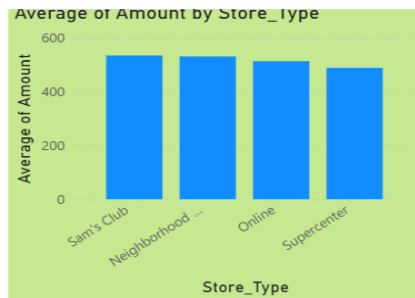
◆ 4. Strengthen Value of Loyalty Tiers

To improve customer retention, Walmart should enhance tier-based benefits such as **bonus rewards, early access to offers, and personalized deals**.

Key Insight — Task 4

The loyalty programme has an engagement crisis. Customers are earning points but not redeeming them. Elite tier customers who earn the most are also the least engaged in redeeming — yet they have the highest churn rate. This suggests the redemption process is too complex or the rewards are not compelling enough to drive re-engagement.

Task 5: Store & Channel Performance



You should now see away line with different values per year — ranging from about 31% to 69%

Store Type	Churn Rate	Share of Total Churn
Supercenter	49.67%	25%
Sam's Club	49.67%	25%
Online	49.67%	25%
Neighborhood Market	49.67%	25%

Store Retention Rate by Opening Year

The line chart with trend line shows Store Retention Rate plotted against Opening Year (1992-2022). The chart reveals a positive correlation — stores opened after 2015 tend to have higher retention rates than stores opened before 2000.

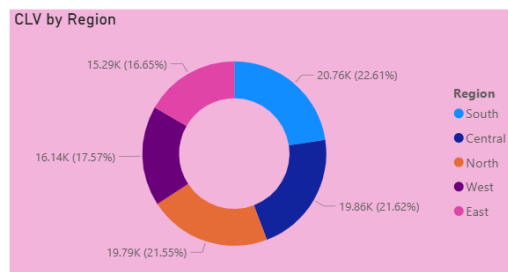
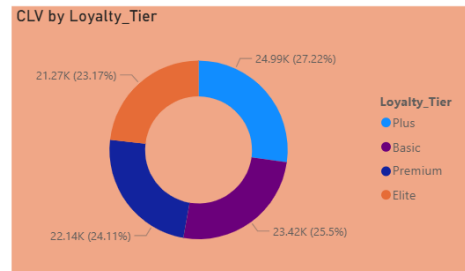
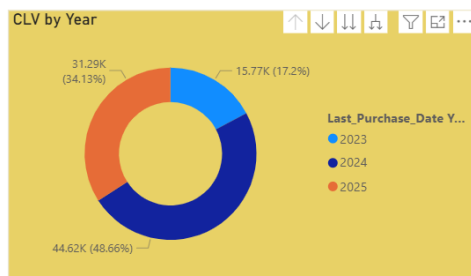
- Stores opened 1992-2000: Average retention rate approximately 46-50%
- Stores opened 2010-2022: Average retention rate approximately 57-69%
- Trend line slopes upward from left to right — confirming positive correlation

Key Insight — Task 5

Newer Walmart stores (opened after 2015) show meaningfully higher retention rates compared to older stores. This may reflect improved store layouts, better technology integration, self-checkout availability, and more modern customer experience design. Walmart should consider renovating or upgrading older stores to bring them to newer standards.

Task 6: Customer Lifetime Value (CLV)

Rs 489	Rs 1.83	Rs 4,767	79
Average CLV	Min CLV	Max CLV	High CLV Customers



6.1 CLV by Year of Last Purchase

CLV is distributed across three years of last purchase activity:

- 2024: Rs 44,620 total CLV (48.66%) — most recent and highest value customers
- 2025: Rs 31,290 total CLV (34.13%)
- 2023: Rs 15,770 total CLV (17.2%) — oldest last purchase, lowest CLV

6.2 CLV by Loyalty Tier

CLV is relatively evenly distributed across loyalty tiers, with Plus tier slightly ahead:

Loyalty Tier	Total CLV	Share	Avg CLV per Customer
Plus	Rs 24,990	27.22%	Highest
Basic	Rs 23,420	25.50%	Second
Premium	Rs 22,140	24.11%	Third
Elite	Rs 21,270	23.17%	Lowest — concerning

6.3 CLV by Region

South region leads in total CLV despite having lower churn, suggesting South customers have higher purchase value per visit:

- South: Rs 20,760 (22.61%) — highest CLV despite moderate purchase frequency
- Central: Rs 19,860 (21.62%)
- North: Rs 19,790 (21.55%)
- West: Rs 16,140 (17.57%)
- East: Rs 15,290 (16.65%) — lowest CLV region

6.4 CLV Segmentation

Customers are divided into High and Low CLV segments based on the average:

- High CLV (above Rs 489/year): 79 customers — these are Walmart's most valuable customers
- Low CLV (below Rs 489/year): 221 customers — majority are low-value customers

Key Insight — Task 6

Elite tier customers have the LOWEST CLV (Rs 21,270 total) despite being the top tier. Combined with their highest churn rate (54.67% from Task 2), this is Walmart's most urgent retention problem. The Elite tier is underperforming on both retention AND lifetime value. Immediate intervention with personalised high-value rewards is critical.

Task 7: Final Dashboard & Executive Summary

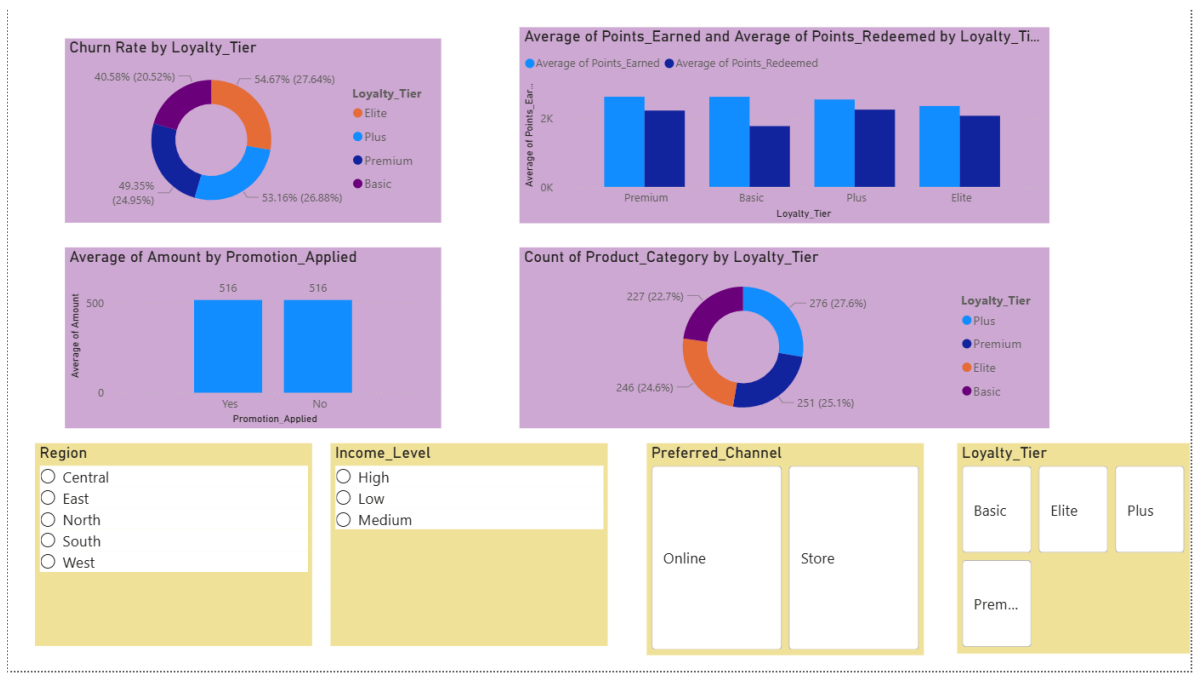
7.1 Dashboard Page 1 — KPI Overview

The KPI page presents the four most critical metrics at a glance, enabling stakeholders to immediately assess the health of customer retention. The funnel chart visually shows the drop from total customers (300) to churned customers (149), and the regional bar chart identifies geographic hot spots.



7.2 Dashboard Page 2 — Loyalty & Promotion

This page deep-dives into loyalty programme effectiveness and promotion impact. The donut chart shows churn distribution by loyalty tier, while the clustered column chart exposes the points redemption gap. The promotion impact chart confirms zero transaction value lift from current promotion strategy.



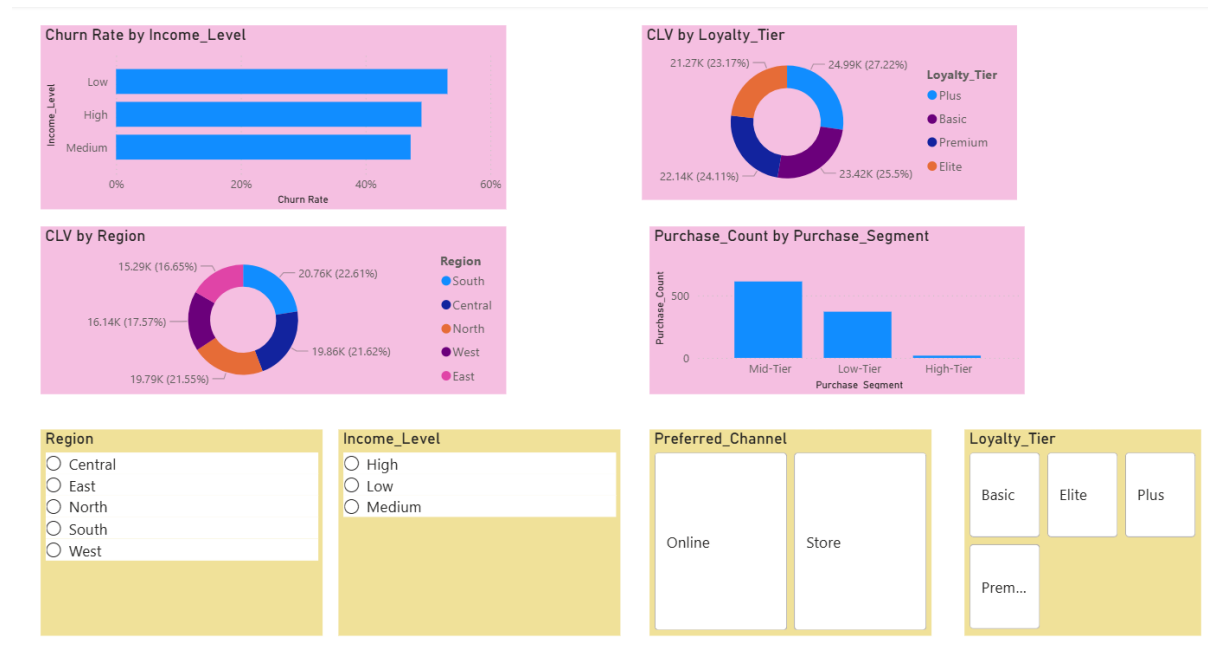
7.3 Dashboard Page 3 — Store & Channel

Store performance analysis reveals that churn is uniform across all store types. The retention-by-opening-year chart shows a positive trend for newer stores. Channel analysis shows online customers churn more than in-store customers.



7.5 Dashboard Page 4 — Customer Segmentation

Segmentation analysis combines CLV, purchase frequency, and demographic data to identify distinct customer groups. Low-Tier purchase segment (57.5% of customers) represents the primary retention challenge



Top 4 Strategic Recommendations

Recommendation 1 — Targeted Retention for Elite Tier Customers

Elite tier customers have the highest churn rate (54.67%) AND the lowest CLV among tiers. Walmart must immediately introduce exclusive high-value benefits for Elite members — personalised deals, dedicated checkout lanes, early sale access, and monthly personalised offers. A 10% improvement in Elite tier retention could recover approximately Rs 2,127 in CLV per customer.

Recommendation 2 — Fix the Promotion Strategy

Promotions are applied to 49% of transactions but generate zero incremental value. Walmart should shift from broad-based discounts to targeted promotions that incentivise larger basket sizes (e.g., 'Spend Rs 800 to unlock double points'). This converts promotions from cost centres into revenue drivers.

Recommendation 3 — Simplify Points Redemption to Re-engage Inactive Customers

All loyalty tiers show a significant gap between points earned and redeemed. Walmart should send monthly 'You have X points expiring' notifications, create a one-click redemption option in the app, and introduce bonus redemption events. This will drive repeat visits from inactive customers and reduce churn.

Recommendation 4 — Convert Low-Tier Buyers to Mid-Tier

57.5% of customers are Low-Tier buyers with only 1-3 purchases. A 'Buy 3 Get Bonus Points' campaign targeting this segment could significantly increase purchase frequency. Converting just 20% of Low-Tier customers to Mid-Tier would add approximately 34 additional repeat buyers to Walmart's active base.

End of Report